[Def.
Given four complex numbers $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$, a map

$$w = T(z) = \frac{a \cdot z + b}{c \cdot z + d}$$

is called a linear fractional transformation.

[Observation for the linear fraction.
The condition $ad - bc \neq 0$ gives that the $T(z)$ is non-constant, assume $ad - bc = 0$,
If $c \neq 0$, then

$$\frac{az+b}{cz+d} = \frac{\frac{a}{c} \cdot (cz+d) + b - \frac{ad}{c}}{cz + d} = \frac{a}{c},$$

If $a \neq 0$, then

$$\frac{az+b}{cz+d} = \frac{az+b}{cz+\frac{bc}{a}} = \frac{a}{c} \cdot \left(\frac{z + \frac{b}{a}}{z + \frac{b}{a}} \right) = \frac{a}{c} \quad \left(z \neq -\frac{b}{a} \right).$$

If $a = 0$, then either $b = 0$ or $c = 0$.
If $b = 0$, then $T \equiv 0$.
If $c = 0$, then $T \equiv \frac{b}{d}$
If $c = 0$, then either $a = 0$ or $d = 0$.
If $a = 0$, then $T \equiv \frac{b}{d}$
If $d = 0$, then T is not defined.

Now, we assume $ad - bc \neq 0$, then on $\mathbb{C} \setminus \{-\frac{d}{c}\}$.

$$w = T(z) = \begin{cases} \frac{a}{c} - \left[\frac{ad - bc}{c^2} \right] \cdot \frac{1}{z + \frac{d}{c}} & \text{if } c \neq 0 \\ \frac{a}{d} z + \frac{b}{d} & \text{if } c = 0, \end{cases}$$

Therefore, the T is injective.

Thm.

Define $\mathbb{H} \overset{\text{def}}{=} \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}$. (That is, the upper plane)

A map $F: \mathbb{H} \to \mathbb{D} : z \mapsto \dfrac{i-z}{i+z}$ is conformal map,

and $G: \mathbb{D} \to \mathbb{H} : w \mapsto i \cdot \dfrac{1-w}{1+w}$ is the inverse of F.

Proof. Given $a+ib \in \mathbb{H}$ $(b>0)$,

$\|(0,1)-(a,b)\|^2 = a^2+(b-1)^2 < \|(0,-1)-(a,b)\|^2 = a^2+(b+1)^2$.

Therefore for all $z \in \mathbb{H}$, $|F(z)|<1$, so the F is well-defined

And, given $u+iv \in \mathbb{D}$ $(u^2+v^2<1)$,

$$\operatorname{Im}(G(w)) = \operatorname{Re}\left[\frac{1-(u+iv)}{1+u+iv}\right] = \operatorname{Re}\left[\frac{(1-u-iv)(1+u-iv)}{(1+u)^2+v^2}\right]$$

$$= \frac{1-u^2-v^2}{(1+u)^2+v^2} > 0$$

Therefore the G is also well-defined. And, these are rational of polynomials, so holomorphic. Now, given $z \in \mathbb{H}$,

$$G(F(z)) = i \cdot \frac{1-\frac{i-z}{i+z}}{1+\frac{i-z}{i+z}} = i \frac{i+z-i+z}{i+z+i-z} = z,$$ and similarly $F \circ G = id$.

So, these are bijection with $G = F^{-1}$ ∎

Observe that for a real number $x \in \mathbb{R}$,

$$F(x) = \frac{i-x}{i+x} = \frac{(i-x)(-i+x)}{1+x^2} = \frac{1+2xi-x^2}{1+x^2} = \frac{1-x^2}{1+x^2} + i \cdot \frac{2x}{1+x^2},$$

and parametrize $x = \tan t$ $\left(-\frac{\pi}{2} < t < \frac{\pi}{2}\right)$, then

$\sin 2a = 2 \cdot \sin a \cdot \cos = \dfrac{2\tan a}{1+\tan^2 a}$ and $\cos 2a = \dfrac{1-\tan^2 a}{1+\tan^2 a}$, so

$F(x) = \cos 2t + i \cdot \sin 2t = e^{2it}$.

Typical Conformal map.

Affine map

$$f : \mathbb{C} \to \mathbb{C} : z \mapsto az + b \quad (a \neq 0)$$

$|a|$ magnitude, rotating $\arg a$.

Möbius transformation

$$T : \overline{\mathbb{C}} \to \overline{\mathbb{C}} : z \mapsto \frac{az+b}{cz+d}, \quad \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} \neq 0.$$

Cayley transform. $\mathbb{H} := \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}$, and $\mathbb{D} = \{z \in \mathbb{C} \mid |z| < 1\}$.

$$C : \mathbb{H} \to \mathbb{D} : z \mapsto \frac{z-i}{z+i}$$

$$C^{-1} : \mathbb{D} \to \mathbb{H} ; w \mapsto i \cdot \frac{1+w}{1-w}.$$

$$\beta : \{z \mid \operatorname{Re} z > 0\} \to \mathbb{D} ; z \mapsto \frac{1+z}{1-z}$$

$$\operatorname{Aut}(\mathbb{D}) = \left\{ z \mapsto e^{i\theta} \cdot \frac{z-a}{1-\overline{a}z} \;\middle|\; |a| < 1, \ \theta \in \mathbb{R} \right\}$$